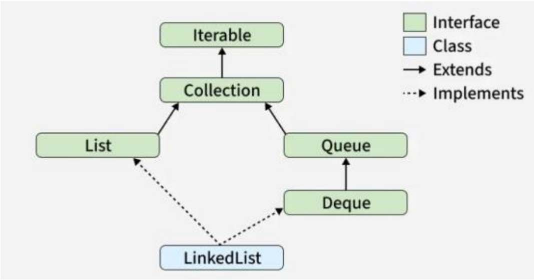
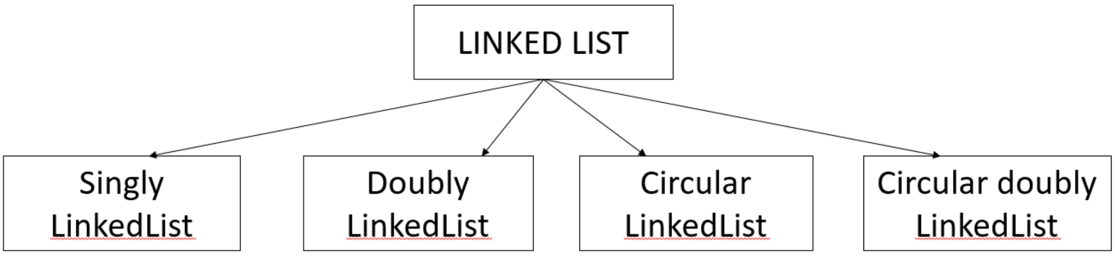
TASK MANAGEMENT SYSTEM

LINKED LIST:



- Linked List is defined as collection of nodes.
- In linked list elements are stored in nodes.
- In linked list each node has two parts (data part & address part).
- Each node has contains data and reference to the next node

TYPES OF LINKED LIST:



Why LinkedList:

- Dynamic size
- Efficient Insertion/Deletion
- No continuous memory required
- Better Memory Utilization
- Easy Implementation of Dynamic Data Structures

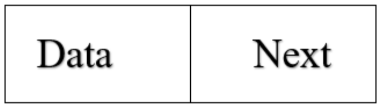
LinkedList Operations:

i. Traversal
ii. Insertion
iii. Deletion

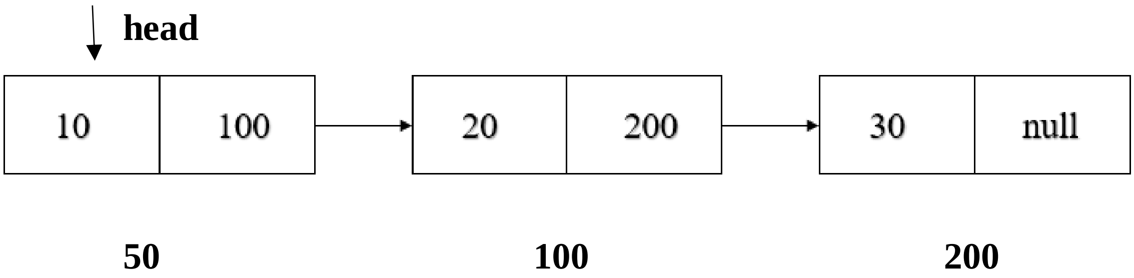
SINGLY LINKEDLIST:

A Singly Linked List is a linear data structure in which each node contains data and a reference to the next node in the sequence. Traversal is possible only in the forward direction.

STRUCTURE OF NODE:



REPRESENTATION OF NODE:



CREATION OF NODE:

```
class Node
{
    int data;
    Node next;
    Node(int data)
   {
        this.data  = data;
        this.next = null;
    }
}
public class Main
{
    public static void main(String[] args)
    {
        Node n1 = new Node(56);
        Node n2 = new Node(78);
     }
}
```

Advantages of Singly Linked List

- Dynamic memory allocation.
- Easy insertion and deletion of nodes.
- Memory is allocated only when required.
- No need to shift elements during insertion or deletion.

Disadvantages of Singly Linked List

- Sequential access only.
- Cannot traverse backward.
- Extra memory is required for storing links

<u>DOUBLY LINKEDLIST:</u>

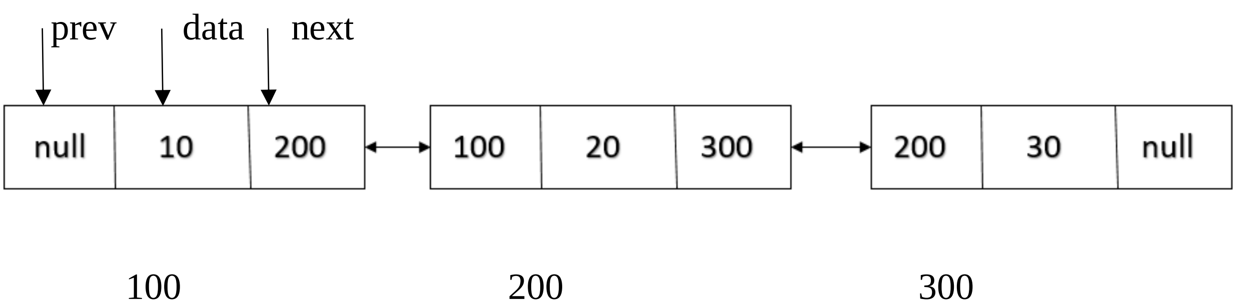
A Doubly Linked List is a type of linked list in which each node contains:

1. Data
2. Address of the Previous Node (prev)
3. Address of the Next Node (next)

Unlike a Singly Linked List, traversal can be performed in both forward and backward directions.

STRUCTURE OF NODE:

REPRESENTATION OD NODE:



CREATION OF NODE:

```
class Node
{
    int data;
    Node prev;
    Node next;
    Node(int data)
    {
        this.data = data;
        this.prev = null;
        this.next = null;
    }
}
public class Main
{
    public static void main(String[] args)
    {
        Node n1 = new Node(56);
        Node n2 = new Node(78);
    }
}
```

ADVANTAGES OF DOUBLY LINKED LIST

- Traversal can be done in both directions.
- Deletion of a node is easier.
- Insertion before or after a node is efficient.

DISADVANTAGES OF DOUBLY LINKED LIST

- Requires extra memory for storing the previous pointer.
- More complex implementation than a Singly Linked List.
- More pointer updates are needed during insertion and deletion.